

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>Member ID</u>	<u>int</u>	<u>system generated</u>
<u>member Name</u>	<u>string</u>	<u>input entered by the member during registration</u>
<u>member Phone</u>	<u>string</u>	<u>input entered by the member</u>
<u>member Email</u>	<u>string</u>	<u>input entered by the member</u>
<u></u>	<u></u>	<u></u>
<u></u>	<u></u>	<u></u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>Book</u>		
Property Name	Data Type	Source Tag
<u>Book ID</u>	<u>int</u>	<u>System Generated</u>
<u>title</u>	<u>string</u>	<u>entered by the librarian when adding a book</u>
<u>author</u>	<u>string</u>	<u>entered by the librarian when adding a book</u>
<u>genre</u>	<u>string</u>	<u>entered by the librarian when adding a book</u>
<u>total Copies</u>	<u>int</u>	<u>entered by the librarian when adding a book</u>
<u>borrow Count</u>	<u>int</u>	<u>calculated from the total number of loan records for this book</u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>Loan</u>		
Property Name	Data Type	Source Tag
<u>loan Id</u>	<u>int</u>	<u>system generated</u>
<u>member Id</u>	<u>int</u>	<u>chosen from the list of registered members</u>
<u>book Id</u>	<u>int</u>	<u>chosen from the list of available books</u>
<u>borrow Date</u>	<u>string</u>	<u>default value ==> automatically set to the current date</u>
<u>return Date</u>	<u>string</u>	<u>default ==> //entered when the member returns the book</u>
<u>due Date</u>	<u>string</u>	<u>calculated from book (calculated as borrow Date + 14 days)</u>
<u>fine Amount</u>	<u>double</u>	<u>calculated form due date</u>

Entity Name: <u>Reservation</u>		
Property Name	Data Type	Source Tag
<u>reservation Id</u>	<u>int</u>	<u>system generated</u>
<u>member Id</u>	<u>int</u>	<u>chosen from the list of registered members</u>
<u>book Id</u>	<u>int</u>	<u>From List chosen from the list of books</u>
<u>resrvation Date</u>	<u>string</u>	<u>default automatically set to the current date</u>
<u>status</u>	<u>string</u>	<u>default value==> automatically set to "Pending" when the reservation is created</u>
<u> </u>	<u> </u>	<u> </u>
<u> </u>	<u> </u>	<u> </u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

Member registers ==> Librarian adds books ==> Member borrows book ==> Member returns book ==> system calculates fine if late ==> system notifies reservations.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	Register Member Service	Creates a new member account in the system.
2	Add Book Service	Allows the librarian to add a new book to the library system.
3	Borrow Book Service	Creates a loan when a member borrows an available book.
4	Return Book Service	Updates the loan when a book is returned and records the return date.
5	Reserve Book Service	Creates a reservation when a book is not available.
6	Calculate Due Date Service	Automatically sets the due date to 14 days after borrowing a book.
7	Calculate Fine Service	Calculates the fine if a book is returned after the due date.
8	Check Book Availability Service	Checks if a book has available copies before borrowing.
9	Notify Reservation Service	Notifies members when their reserved book becomes available
10	Generate Reports Service	Generates reports such as most borrowed books and overdue loans

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
guestId	int	System Generated
guestName	string	input //guest type's their name during registration
guestPhone	string	Input //guest enters phone number
guestEmail	string	input// guest enters email
guestNationality	string	input
registrationDate	string	default value

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	System Generated
roomNumber	string	input /// entered by receptionist
roomType	string	from list // receptionists select Single, Double, Suite
floorNumber	int	input entered by staff
nightlyPrice	decimal	Calculated based on room type rules in system
isAvailable	bool	Default Value starts as true when room is added

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	System Generated
guestId	int	From List (selected from registered guests list)
roomId	int	From List (selected from available rooms list)
checkInDate	string	Default Value (set automatically when booking is created)
checkOutDate	string	Default Value (set when guest checks out)
numberOfNights	int	Calculated (difference between check-in and check-out)
totalPrice	decimal	Calculated (nights × room price)
status	string	Default Value (starts as "Booked")

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System Generated
bookingId	int	From List (bookings)
guestId	int	From List (bookings)
roomId	int	
actualCheckIn	string	Default Value (system time when guest arrives)
actualCheckOut	string	Default Value (system time when guest leaves)
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Guest registers ==> Guest profile created
Hotel staff adds rooms ==> Room created
Guest books room ==> Booking created + room becomes unavailable
Guest checks in ==> CheckInRecord created
Guest checks out ==> system calculates price + room becomes available
Booking cancelled ==> room becomes available again

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuestService	Creates a new guest account
2	AddRoomService	Adds a new room to the system
3	CreateBookingService	Creates a booking for guest and room
4	CheckRoomAvailabilityService	Checks if room is available
5	CalculateTotalPriceService	Calculates total stay price
6	CheckInService	Creates check-in record
7	CheckOutService	Handles guest checkout

#	Service Name	What it does
8	<u>CancelBookingService</u>	<u>Cancels booking and frees room</u>
9	<u>UpdateRoomStatusService</u>	<u>Updates room availability</u>